Covariate-Dependent Vector Auto-Regresion and Universal Structural Equation Models for Explainable Idiographic Structure in Time Series

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Abstract

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1 Introduction

- There is a growing availability of longitudinal data, which allows for an increase in the breath and depth of research regarding how specific psychometric measurements influence behavior, well-being and daily progress.

- Often, this data takes the form of a set of time-series, each associated with an individual patient. When modeling, we must take into account both the temporal dependency of the data (prior psychological states are likely to influence future states), and the dependencies among the psychometric measurements.

- Many methods have been proposed for the estimation of such dependencies such as sparse vector auto-regression models [1], dynamic factor analysis [8], and the universal structural equation modeling (USEM) [5].

- In psychometric research, it is vital to recognize the effect that individual characteristics of each subject can have in the overall fit of the model and interpretation. From this, a variety of works have focused on the estimation of a common (or nomothetic) structure across all individuals, as well as individual (or idiographic) structure that is exclusive to each subject. The GIMME method [4] performs the estimation of the network of psychometric relationships via the estimation of multiple USEM model, with a variable selection procedure that estimates model overlaps. More recently, the MultiVAR method [3] estimates the individual and common structure through a penalized optimization approach. Further extensions to consider the potential structure of subgroups has been considered for both approaches [6; 2].

- Often, in psychometric or MRI studies, our subject-specific time series data is accompanied by additional clinical data. The incorporation of this additional information may provide further insight into behavior, but its use has been largely ignored.

- One important omission from these approaches is to explain *why* each individual has a particular idiographic structure. While the common network has a straightforward interpretation, the individual structures are not necessarily easy to interpret or process. Therefore, the output of GIMME or MultiVAR can only measure the existence of this idiographic structure, without explaining in any way the origin or motivation of potential underlying patterns.

- In the current writeup, we propose a method for estimating the nomothetic and idiographic structure of multi-subject time series, where the time dependence of each individual depends on a set of underlying covariates. This way, the idiographic effects are not simply unexplained individual structure, but are instead linked to the patients' individual characteristics. This allows for an actual interpretation of the idiographic structure.

- **Main advantages**: while other methods allow for modeling nomothetic and idiographic structure, the motivation for why such variations occur remains largely unknown. The explanation is simply: there is variation across individuals. This method connects the individual variation to underlying features, allowing us to determine the nature of the nomothetic structure, and to provide potential explanations for the origin of the idiographic structure. This can then be used as useful knowledge for further research or downstream statistical analysis.

2 Description of the Model

- Consider we have a study with N subjects. For each subject $1 \leq k \leq N$, we have access to two data modes: (i) a p-dimensional vector of covariates $\boldsymbol{X}_k \in \mathbb{R}^p$, and (ii) a d-dimensional time series of length T_k, given as $\mathbf{Y}_k = \{\boldsymbol{Y}_t^{(k)} \in \mathbb{R}^d \,:\, 1 \leq t \leq T_k\}$.

- In the usual GIMME or MultiVAR frameworks [4; 3], the aim is to model the time-series with a common+individual decomposition, usually of the form:
$$\boldsymbol{Y}_t^{(k)} = \sum_{\ell=1}^{q}(\Psi_\ell^c + \Psi_\ell^{(k)})\boldsymbol{Y}_{t-\ell}^{(k)} + \boldsymbol{e}_t^{(k)}. \qquad \text{(MultiVAR);}$$
$$\boldsymbol{Y}_t^{(k)} = (\mathrm{A}^c + \mathrm{A}^{(k)})\boldsymbol{Y}_t^{(k)} + \sum_{\ell=1}^{q}(\Phi_\ell^c + \Phi_\ell^{(k)})\boldsymbol{Y}_{t-\ell}^{(k)} + \varepsilon_t^{(k)}. \qquad \text{(GIMME).}$$

- While these approaches are useful at extracting common and individual structure, they miss the opportunity to exploit the potentially useful additional information contained in $\{\boldsymbol{Y}_k\}_{k=1}^N$.

- To exploit it, we assume that the time-series model coefficients depend on the values of Y. We consider the following models:

$$\boldsymbol{Y}_t^{(k)} = \sum_{\ell=1}^{q} \Psi_\ell(\boldsymbol{X}_k)\boldsymbol{Y}_{t-\ell}^{(k)} + \boldsymbol{e}_t^{(k)}. \qquad \text{(MultiVAR);} \qquad (1)$$

$$\boldsymbol{Y}_t^{(k)} = \mathrm{A}(\boldsymbol{X}_k)\boldsymbol{Y}_t^{(k)} + \sum_{\ell=1}^{q} \Phi_\ell(\boldsymbol{X}_k)\boldsymbol{Y}_{t-\ell}^{(k)} + \varepsilon_t^{(k)}. \qquad \text{(GIMME).} \qquad (2)$$

  Notice that, instead of being fixed, the set of coefficients $\Psi_\ell(\boldsymbol{X}_k), \mathrm{A}(\boldsymbol{X}_k), \Phi_\ell(\boldsymbol{X}_k)$ now depend on the individual covariate data. From this, the individual structure of each cannot be arbitrary, but it now depends on the specific features $\boldsymbol{X}_k$.

- As a first approach, lets assume that these parameters have an *additive* relationship to the underlying covariates. For a vector $\boldsymbol{x} = (x_1, x_2, \dots, x_p)^\top \in \mathbb{R}^p$

$$\Psi_\ell(\boldsymbol{x}) = \Psi_{\ell 0} + x_1\Psi_{\ell 1} + x_2\Psi_{\ell 2} + \ldots + x_p\Psi_{\ell p};$$

$$\mathrm{A}(\boldsymbol{x}) = \mathrm{A}_0 + x_1\mathrm{A}_1 + x_2\mathrm{A}_2 + \ldots + x_p\mathrm{A}_p;$$

$$\Phi_\ell(\boldsymbol{x}) = \Phi_{\ell 0} + x_1\Phi_{\ell 1} + x_2\Phi_{\ell 2} + \ldots + x_p\Phi_{\ell p}.$$

  Here, the matrices $\{\Psi_{\ell 0}\}_{\ell=1}^q, \mathrm{A}_0, \{\Phi_{\ell 0}\}_{\ell=1}^q$ represent the structure that is common (nomothetic) among all subjects. The remaining parameters are affected by $\boldsymbol{x}$, so they are dependent on the individual characteristics of each subject.

- **Example:** We consider a VAR(1) model, where there is a single lagged relationship, *i.e.* $q = 1$, and two subject-level covariates $p = 2$. In that case, given that $\boldsymbol{X}_k = (X_{k1}, X_{k2})^\top$ our equations simplify to:

$$\begin{aligned}\boldsymbol{Y}_t^{(k)} &= \Psi(\boldsymbol{X}_k)\boldsymbol{Y}_{t-1}^{(k)} + \boldsymbol{e}_t^{(k)} \\ &= [\Psi_0 + X_{k1}\Psi_1 + X_{k2}\Psi_2] \cdot \boldsymbol{Y}_{t-1}^{(k)} + \boldsymbol{e}_t^{(k)}\end{aligned}$$

  Notice that each subject k will have their own VAR structure. This structure will incorporate a common structure shared across all subjects $\Psi_0 \in \mathbb{R}^{d\times d}$. It also has individual structure. Now, instead of having this individual structure vary arbitrarily, it has the form $Y_{k1}\Psi_1 + Y_{k2}\Psi_2$. Therefore, it depends on known information $Y_k = (Y_{k1}, Y_{k2})'$ about each subject.

- In Figure 2, we provide an example of how common and individual effects in a VAR time series model can relate to underlying subject-level covariates. Here, the dimen-

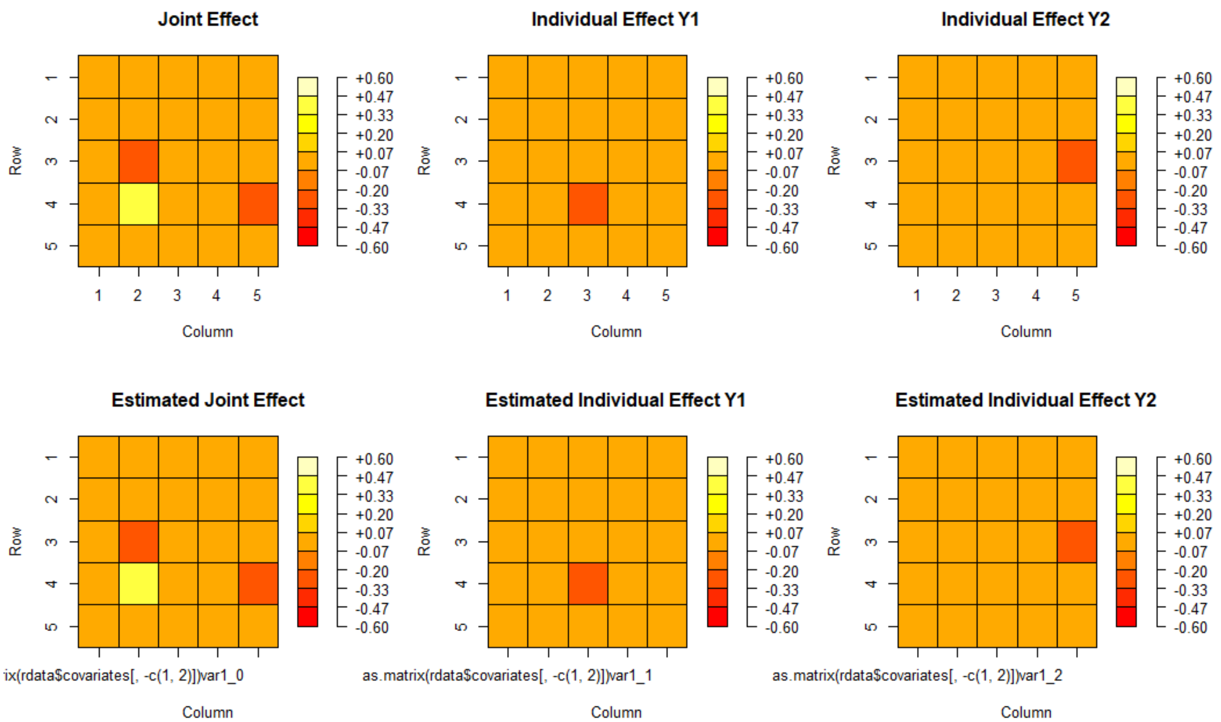

Figure 1: Illustration of the true and estimated Regression VAR parameters. For very large sample sizes, the traditional OLS method is successful. It would now be interesting to explore how regularization or model selection can help in higher dimension.

sion of our time series is $d = 5$, the number of subject-level covariates is $p = 2$, and the lag in the model is $q = 1$. We generate data for $N = 50$ subjects, all with a total of $T = 100$ time points.

- For this same example, we performed a traditional LS fitting, to see if we could successfully recover the common/individual structure of the time series. As we see in Figure 1, our estimation for very large sample sizes is successful.

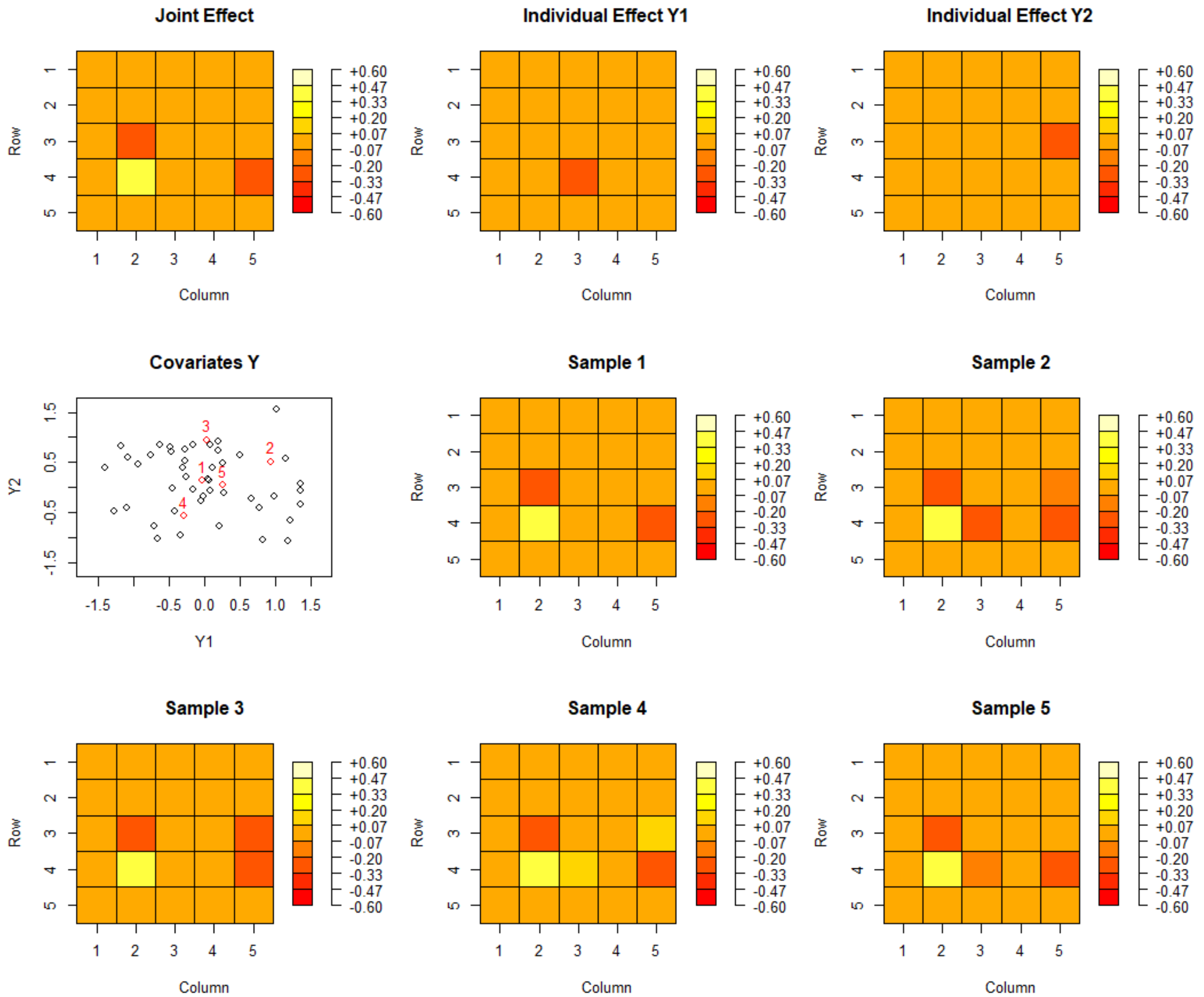

Figure 2: Illustration of the Regression VAR framework with 2 covariate-dependent effects. Top-panels: visualization of the joint effects that are common across all subjects. We also visualize the effects that are individual to variables X_1 and X_2. Center-left panel: Visualization of the covariate information for $N = 50$ subjects. We highlight the covariate information of the first 5 subjects. Remaining panels: Visualization of the effects for the first 5 subjects. We observe that, as subject 1 has nearly zero for both X_1 and X_2, the effects are mostly just the joint effects. Subject 3 has a high X_2 covariate, but low X_1 covariate, and therefore its effects are a mixture of the joint and X_2 effects. Subject 2 has large covariates for both X_1 and X_2, so its time series effects are a mixture of all.

3 Fitting the RVAR Model with LASSO Penalty

In this section, we describe the fitting of the RVAR model of the form (1), with 1-lag relationships, *i.e.* $q = 1$. Assuming that, for each individual $k = 1, ..., N$, we can capture the VAR relationships in matrix form as:

$$\underbrace{\begin{bmatrix} (\boldsymbol{Y}^k_{T_k})' \\ (\boldsymbol{Y}^k_{T_k-1})' \\ \vdots \\ (\boldsymbol{Y}^k_2)' \end{bmatrix}}_{\mathbf{Z}^k} = \underbrace{\begin{bmatrix} (\boldsymbol{Y}^k_{T_k-1})' & X_{k1}(\boldsymbol{Y}^k_{T_k-1})' & X_{k2}(\boldsymbol{Y}^k_{T_k-1})' & \dots & X_{kp}(\boldsymbol{Y}^k_{T_k-1})' \\ (\boldsymbol{Y}^k_{T_k-2})' & X_{k1}(\boldsymbol{Y}^k_{T_k-2})' & X_{k2}(\boldsymbol{Y}^k_{T_k-2})' & \dots & X_{kp}(\boldsymbol{Y}^k_{T_k-2})' \\ \vdots & \vdots & \ddots & \vdots & \\ (\boldsymbol{Y}^k_1)' & X_{k1}(\boldsymbol{Y}^k_1)' & X_{k2}(\boldsymbol{Y}^k_1)' & \dots & X_{kp}(\boldsymbol{Y}^k_1)' \end{bmatrix}}_{\mathbf{W}^k} \underbrace{\begin{bmatrix} \Psi'_0 \\ \Psi'_1 \\ \vdots \\ \Psi'_p \end{bmatrix}}_{\Psi} + \underbrace{\begin{bmatrix} (\varepsilon^k_T)' \\ (\varepsilon^k_{T-1})' \\ \vdots \\ (\varepsilon^k_{T-q+1})' \end{bmatrix}}_{\mathbf{E}^k}.$$

This can be simplified to matrix form as

$$\mathbf{Z}^k = \mathbf{W}^k \boldsymbol{\Psi} + \mathbf{E}^k \tag{3}$$

The equation (3) represents the VAR relationships for individual k. Then, we can model the behavior for all individuals via the equation

$$\underbrace{\begin{bmatrix} \mathbf{Z}^1 \\ \mathbf{Z}^2 \\ \vdots \\ \mathbf{Z}^N \end{bmatrix}}_{\mathbf{Z}} = \underbrace{\begin{bmatrix} \mathbf{W}^1 \\ \mathbf{W}^2 \\ \vdots \\ \mathbf{W}^N \end{bmatrix}}_{\mathbf{W}} \boldsymbol{\Psi} + \underbrace{\begin{bmatrix} \mathbf{E}^1 \\ \mathbf{E}^2 \\ \vdots \\ \mathbf{E}^N \end{bmatrix}}_{\mathbf{E}}, \tag{4}$$

Thus, we can summarize the VAR equation model for all individuals simultaneously with the reduced matrix equation:

$$\mathbf{Z} = \mathbf{W}\boldsymbol{\Psi} + \boldsymbol{E}, \tag{5}$$

where $\mathbf{Z} \in \mathbb{R}^{(T-N)\times d}$, $T = \sum_k T_k$, $\mathbf{W} \in \mathbb{R}^{(T-N)\times(d(p+1))}$ and $\mathbf{E} \in \mathbb{R}^{(T-N)\times d}$.

To solve for the parameter $\boldsymbol{\Psi} \in \mathbb{R}^{d(p+1)\times d}$, we propose the following optimization problem,

$$\widehat{\boldsymbol{\Psi}} := \widehat{\boldsymbol{\Psi}}(\lambda_1, \lambda_2) = \underset{\boldsymbol{\Psi}\in\mathbb{R}^{d(p+1)\times d}}{\operatorname{argmin}} |||\mathbf{Z} - \mathbf{W}\boldsymbol{\Psi}|||^2_F + \lambda_1 |||\Psi_0|||_1 + \lambda_2 |||[\Psi_1\ \Psi_2\ \dots\ \Psi_p]|||_1, \tag{6}$$

where the hyperparameter λ_1 serves to penalize the common structure of all VAR models, and λ_2 penalizes the X covariate dependent portion of the model. We can interpret (6) as a matrix regression problem. Notice that the optimization problem (6) is equivalent to solving the separate column-wise optimization problems

$$\widehat{\boldsymbol{\Psi}}^\ell(\lambda_1, \lambda_2) = \underset{\boldsymbol{\Psi}^{(\ell)}\in\mathbb{R}^{d(p+1)}}{\operatorname{argmin}} \left|\left|\left|\mathbf{Z}_{\cdot\ell} - \mathbf{W}\boldsymbol{\Psi}^{(\ell)}\right|\right|\right|^2_F + \lambda_1 \|[\boldsymbol{\Psi}^{(\ell)}]_{1:d}\|_1 + \lambda_2 \|[\boldsymbol{\Psi}^{(\ell)}]_{(d+1):(dp)}\|_1. \tag{7}$$

When solving the individual optimization problems (3.1), we can set the estimated parameter $\widehat{\mathbf{\Psi}}(\lambda_1, \lambda_2) := \left[\widehat{\mathbf{\Psi}}^1\, \widehat{\mathbf{\Psi}}^2\, \ldots\, \widehat{\mathbf{\Psi}}^d\right](\lambda_1, \lambda_2) \in \mathbb{R}^{d(p+1)\times d}$. Figure 1 in the previous section was derived with $\lambda_1 = \lambda_2 = 0$.

The current implementation only allows for a single lagged relationship, *i.e.* $q = 1$. We aim to create a more general implementation with lag $q > 1$ in the future. Alternatively, we can further vectorize the representation (5) into a single-response linear regression of the form

$$\mathbf{vec}(\boldsymbol{Z}) = (I_d \otimes \mathbf{W}) \cdot \mathbf{vec}(\Psi) + \mathbf{vec}(\mathbf{E})$$

where $\mathbf{vec}(\mathbf{A}) = [A_{\cdot 1}^\top, A_{\cdot 2}^\top, \ldots A_{\cdot n}^\top]^\top \in \mathbb{R}^{ab}$ and $\mathbf{A}\otimes\mathbf{B} = [A_{ij} \cdot \mathbf{B}]_{i,j} \in \mathbb{R}^{ac\times bd}$ for any $\mathbf{A} \in \mathbb{R}^{a\times b}$ and $\mathbf{B} \in \mathbb{R}^{c\times d}$. Thus, solving (3.1) is equivalent to solving a weighted-penalty version of the LASSO add LASSO/glmnet ref.

3.1 Adaptive CD-VAR

To improve upon the performance of our original estimator, we propose a refinement of our CD-VAR estimator through an adaptive approach. For this, let $\widehat{\mathbf{\Psi}}$ be an initial estimator of the CD-VAR model. Then, given a constant $\gamma > 0$ we define the adaptively-weighted ℓ_1 penalty as:

$$P_{ada}(\mathbf{\Psi}) = \sum_{\ell=1}^{p} \sum_{i,j=1}^{d} |\Psi_{ij}^{(\ell)}|^{-\gamma} \cdot |\mathbf{\Psi}_{ij}^{(w)}| \tag{8}$$

Then, for $\lambda > 0$, our adaptive CD-VAR estimator is given as a solution to the optimization problem

$$\widehat{\mathbf{\Psi}}_{ada}(\lambda_a) = \underset{\mathbf{\Psi}\in\mathbb{R}^{d\times(d(p+1))}}{\operatorname{argmin}} \; \|\mathbf{Z} - \mathbf{W}\mathbf{\Psi}\|_F^2 + \lambda P_{ad}(\mathbf{\Psi}).$$

In following sections, we explore how the adaptive approach can help improve the edge-recovery performance.

3.2 Tuning Parameter Selection

One of the main concerns when fitting a penalized model is to determine the best procedure for selecting tuning parameters. In particular, due to the dependence present in time series data, direct sample splitting is not feasible, leading to sophisticated procedures of data splitting to obtain a reasonable cross-validation procedure. For example add references, and what they do.

3.2.1 Cross-Validation

Contrary to other methods like GIMME and Multi-VAR, one can generate "predictive model coefficients": with a model $\hat{\mathbf{\Psi}} \in \mathbb{R}^{d\times(p(d+1))}$ trained on the data of N individuals $\{(\boldsymbol{X}_k, \mathbf{Y}^{(k)})\}_{k=1}^{N}$, we may provide an estimate of the coefficients of an out-of-sample individual $\{(\boldsymbol{X}_{N+1}, \mathbf{Y}^{(N+1)})$ via $\hat{\Phi}_{N+1} = \hat{\Psi}_0 + \sum_{i=1}^{p} X_{ki}\hat{\Psi}_i$.

In our particular modeling, due to the independence across individuals, it is possible for us to perform cross validation as follows. Given a number of subjects N and number of folds $K < N$, we divide the set of individuals $\{1, 2, \ldots, N\}$into K groups $\mathcal{G}_1, \ldots, \mathcal{G}_K$. Then, for each $k \in [K]$, the k-th fold consist of the union of all the data corresponding to the individuals in $\mathcal{G}_k$.

While simple, this approach for cross-validation cannot be used in usual VAR models, or even the Multi-VAR and GIMME methods. The main reason is that our model assumes a structural relationship between the exogenous covariates and the individual time-series model. Therefore, the training on a set of individuals $\mathcal{G} \subset [N]$ can be used to infer the time-series coefficients for a new individual $i \in [N]\backslash\mathcal{G}$. In contrast, methods like GIMME, Multi-VAR and their extensions must train individual coefficients for each individual separately. Therefore, fitting the time-series models for a subset of individuals $\mathcal{G} \subset [N]$ cannot be exploited to determine the coefficients of a new individual $i \in [N] \setminus \mathcal{G}$, or at least not directly.

On our case, since we assume that there is an underlying additive structure that connects all the individual time-series models, it is possible to train both the invidiual and common structure simultaneously.

3.2.2 BIC

In addition to cross-validation, we alternatively propose the fitting of our model by selecting the tuning parameter that minimizes the Bayesian information criteria (BIC). Given $\lambda_1, \lambda_2 > 0$, we define,

$$\mathrm{BIC}(\lambda_1, \lambda_2) = \left\|\left|\mathbf{Z} - \mathbf{W}\widehat{\mathbf{\Psi}}(\lambda_1, \lambda_2)\right\|\right|_F^2 + \left\|\left|\widehat{\mathbf{\Psi}}(\lambda_1, \lambda_2)\right\|\right|_0 \log(T - N). \tag{9}$$

Through numerical results, we have verified that BIC is a measure that leads to more sparse estimates of the matrix $\mathbf{\Psi}$ when compared to the estimate obtained through cross-validation.

4 Fitting RUSEM

The implementation of the RVAR model described in Section 3 is simply a generalization of the optimization problem considered in the Fisher et al. [3]. Notice that the RVAR focuses the estimation on the lagged effects $\mathbf{\Psi}$. For the GIMME model, we are also interested in recovering contemporaneous effects. The GIMME model cannot easily be extended to the regression setting. Therefore, we need to explore a different procedure for recovering the USEM structure in the regression setting. This will require thought and research...

I explored the original GIMME reference Gates and Molenaar [4], which points towards the modification index described in Sörbom [7]. I am still trying to understand how the GIMME adapts the estimation of USEM described in Sörbom [7] to scenarios with time dependence. This will be the next task.

5 Simulation Study for R-VAR

To explore the performance of our proposed RVAR model, and to compare to other methods found in the literature, we perform a numerical simulation study.

5.1 Simulation Setup

We consider multivariate time series over $d = 10, 20, 30$ variables of interest. We consider scenarios with $N = 10, 20$ subjects, measured over $T = 30, 50, 100$ time points. For each subject $k \in \{1, 2, \dots, N\}$, we simulate a vector of $p = 2, 5$ time-invariant covariates according to the mixture distribution $\boldsymbol{X}_k = (X_{k1}, X_{k2}, \dots, X_{kp}) \overset{i.i.d.}{\sim} 0.25 \cdot \delta_0 + 0.75 \cdot \textbf{Uniform}([-1, -0.5] \cup [0.5, 1])$. Then, we simulate a lag-1 time series dataset for the subject $k \in \{1, 2, \dots, N\}$ according to the covariate-dependent VAR model (1) with parameter $\mathbf{\Psi}(\boldsymbol{X}_i)$, where

$$\mathbf{\Psi}(\boldsymbol{x}) = \Psi_0 + x_1\Psi_1 + x_2\Psi_2 + \dots + x_p\Psi_p.$$

Finally, the common effect matrix $\Psi_0 \in \mathbb{R}^{d\times d}$ and the matrix of covariate-dependent effects $\Psi_1, \ldots, \Psi_p \in \mathbb{R}^{d\times d}$ are generated as follows. First, we select the proportion of common non-zero entries η_c, a proportion of covariate-dependent non-zero entries η_i. We generate Ψ_0 by selecting at random $d^2\eta_c$ entries to follow the distribution $\textbf{Uniform}([-0.9, -0.3]\cup[0.3, 0.9])$. To generate Ψ_k, we select $d^2\eta_i$ to follow $\textbf{Uniform}([-0.9, -0.3] \cup [0.3, 0.9])$. In our simulations, we consider $(\eta_c, \eta_i) = \{(0.075, 0.025), (0.025, 0.075)\}$. Finally, we generate time series for the k-th subject $\{\boldsymbol{Y}_1^{(k)}, \boldsymbol{Y}_2^{(k)}, \ldots, \boldsymbol{Y}_{T+400}^{(k)}\}$ via the model (1), with $\boldsymbol{e}_1^{(k)}, \boldsymbol{e}_2^{(k)}, \ldots, \boldsymbol{e}_T^{(k)} \overset{i.i.d.}{\sim} N_d(\mathbf{0}, (0.1)\mathrm{I}_d)$, and discard the first 400 time steps.

We compare our proposed sparse CD-VAR method with methods of multi-subject vector auto-regressive models found in the literature. In particular, we consider the estimation of the time series via the individual VAR model [1], as well as the Multi-VAR method in their original and adaptive versions [3]. For the selection of the tuning parameters of the CD-VAR method, we consider training based on cross-validation, as well as based on our proposed BIC measure. Furthermore, we fit our method with both a direct and two-step adaptive approach, as described in Section 3.1.

We compare the methods in terms of several measurements of performance. First, we consider the mean root aggregated error (MRAE), and mean absolute aggregated error (MAAE), given by,

$$MRAE = \frac{1}{N}\sum_{k=1}^{N}\sum_{\ell=1}^{p}\left\|\hat{\Psi}_\ell^k - \Psi_\ell^k\right\|_F;$$
$$MAAE = \frac{1}{N}\sum_{k=1}^{N}\sum_{\ell=1}^{p}\left\|\hat{\Psi}_\ell^k - \Psi_\ell^k\right\|_1;$$

5.2 Simulation Results

The results of the MRAE for $d = 30$ are presented in Figure 3. The best performance for all scenarios is obtained by the adaptive CV-SCDVAR model, closely followed by the original CV-SCDVAR model and the adaptive BIC-SCDVAR model. This findings are consistent across choices of number of explanatory covariates $p = 2, 5$, number of subjects $N = 10, 20$, common entries $p_c = 0.025, 0.075$ and number of available time points T.

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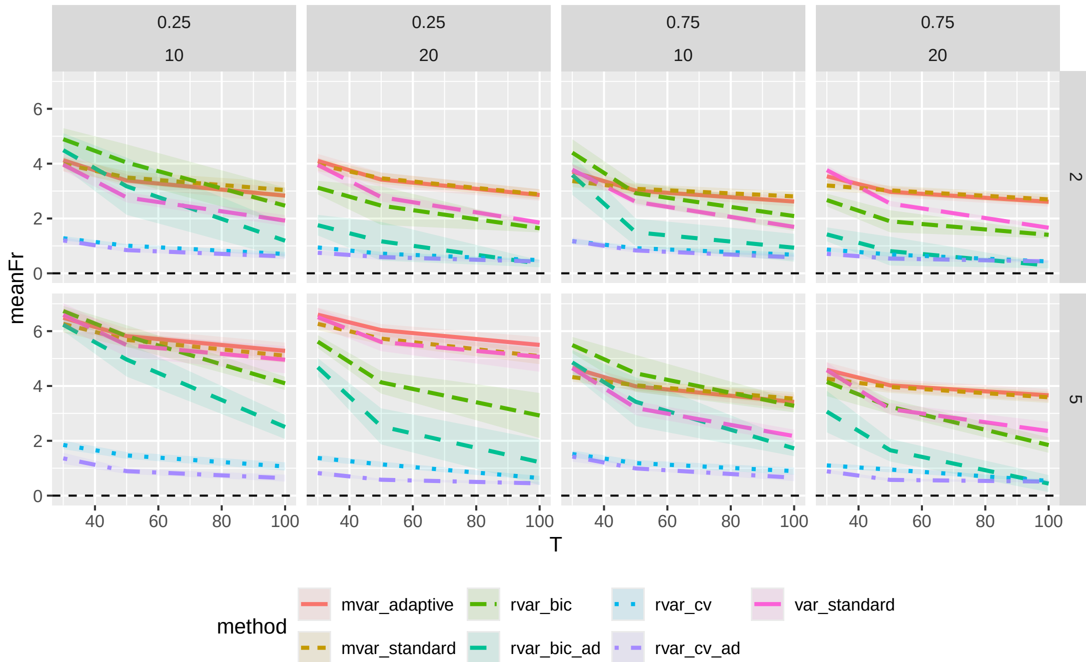

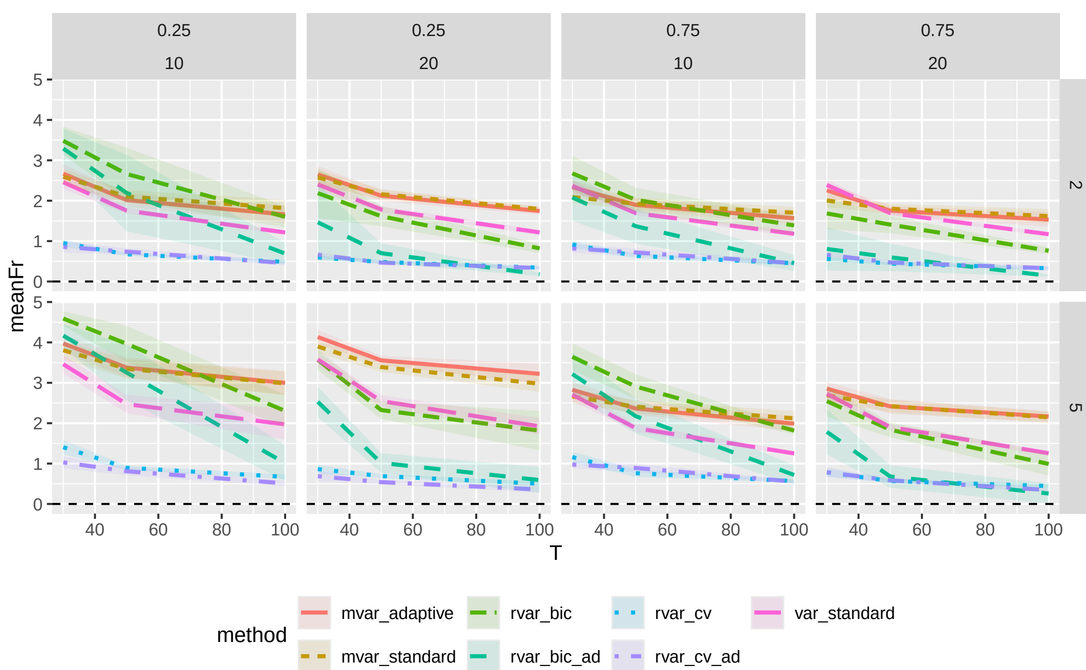

Figure 3: This is the caption for the inserted PDF page.